

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL AS CHEMISTRY (9620)

Unit 2: Organic 1 and Physical 1

Thursday 9 January 2025 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 70.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



Answer **all** questions in the spaces provided.

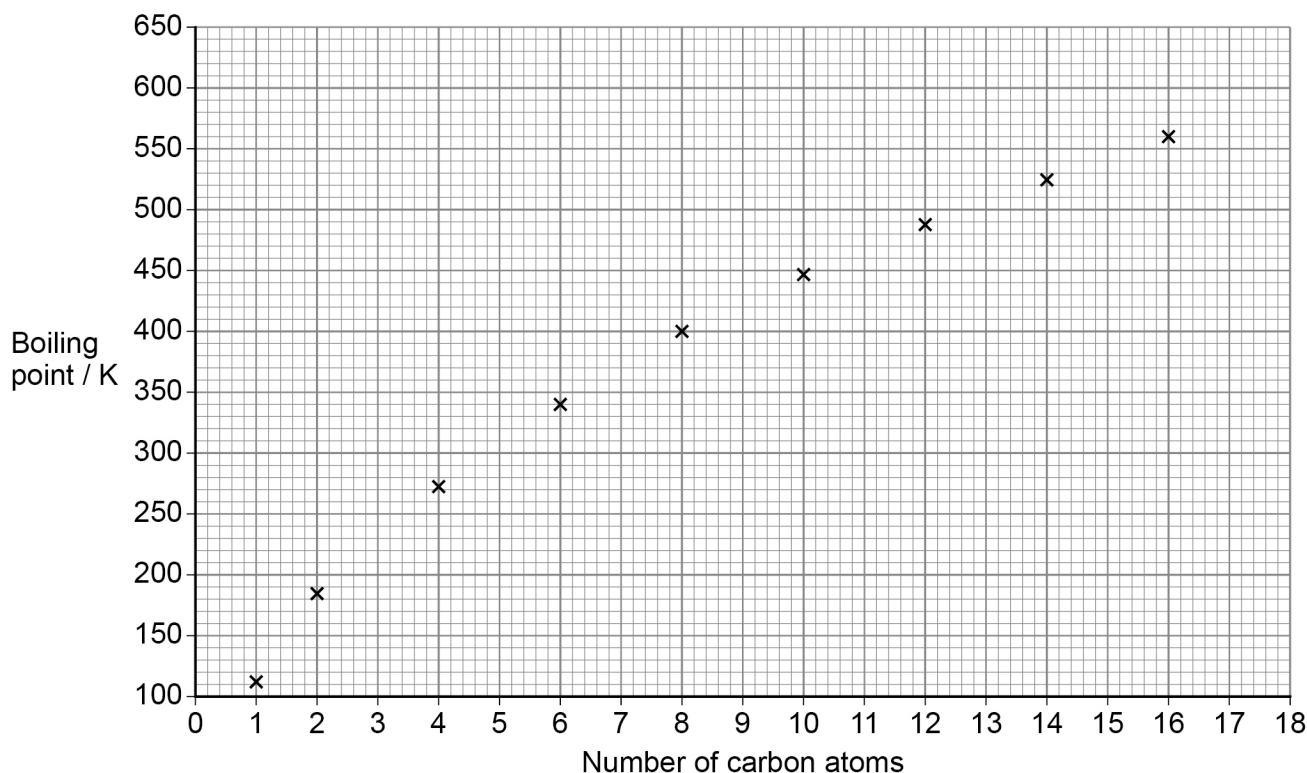
0 1

This question is about hydrocarbons.

Figure 1 shows a graph of the boiling points of some straight chain alkanes.

Octadecane ($C_{18}H_{38}$) is a straight chain alkane.

Figure 1



0 1 . 1

Draw a line of best fit on **Figure 1**.

Continue the line to determine a value for the boiling point of octadecane.

[2 marks]

Boiling point of octadecane _____ K

0 1 . 2

Name a process that can be used to separate octadecane from a mixture of alkanes.

[1 mark]



Alkanes are burned in air in internal combustion engines of cars.

0 1 . 3

Identify **two** toxic gases that are formed in internal combustion engines of cars.

Write **one** equation to show how **both** toxic gases can be removed by a catalytic converter in these cars.

Identify the catalyst used in the catalytic converter.

[3 marks]

Toxic gases _____

Equation

Catalyst _____

0 1 . 4

Write an equation for the complete combustion of octadecane ($C_{18}H_{38}$)

[1 mark]

0 1 . 5

Write an equation for the cracking of one molecule of octadecane to give one molecule of decane ($C_{10}H_{22}$), one molecule of pentene and one other product.

[1 mark]

8

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0	2
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This question is about isomers.

0	2	.	1
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Compounds **D** and **E** have the molecular formula C_4H_{10}
D has a higher boiling point than **E**.

Draw the skeletal formula of **D** and the skeletal formula of **E**.

[2 marks]

D

E

0	2	.	2
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Compounds **F** and **G** have the molecular formula C_4H_8
Bromine water turns colourless when added to **F** but there is no visible change with **G**.

Draw a structure for **F** and a structure for **G**.

[2 marks]

F

G

0	2	.	3
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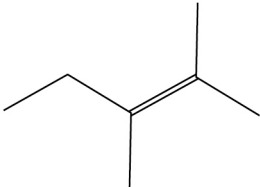
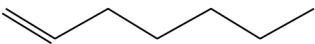
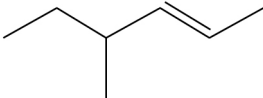
Define stereoisomerism

[1 mark]



0 2 . 4 Table 1 shows the skeletal formulas of three isomeric alkenes.

Table 1

A	B	C
		

Tick (✓) **one** box to choose the alkene, **A**, **B** or **C**, that can show *E-Z* isomerism.

[1 mark]

A

☐

B

☐

C

☐

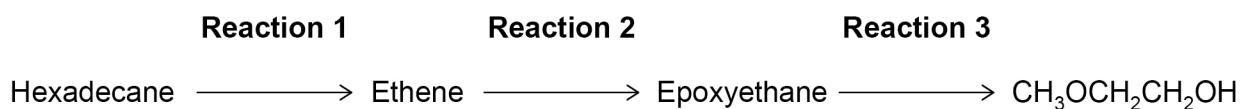
0 2 . 5 Draw the structure of Z-5-methylhex-2-ene.

[1 mark]

0 3

A reaction sequence starting from hexadecane ($\text{C}_{16}\text{H}_{34}$) is shown in **Figure 2**.

Figure 2



0 3 . 1

In **Reaction 1** the only products are octane (C_8H_{18}) and ethene.

Write an equation using molecular formula for **Reaction 1**.

[1 mark]

0 3 . 2

Write an equation for **Reaction 2**.

Show the displayed formula of epoxyethane in your answer.

[1 mark]

0 3 . 3

Name the catalysts used in **Reaction 1** and in **Reaction 2**.

[2 marks]

Catalyst in **Reaction 1** _____

Catalyst in **Reaction 2** _____

0 3 . 4

Identify the substance that reacts with epoxyethane in **Reaction 3**.

[1 mark]



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ANSWER IN THE SPACES PROVIDED**

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0 4

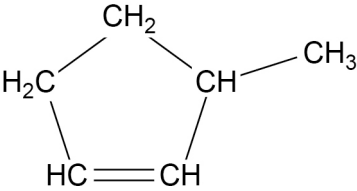
This question is about polymers.

0 4 . 1

Complete **Table 2** to show the

- structure of the monomer used to form polymer **K**
- repeating unit of the polymer formed from monomer **L**

[2 marks]**Table 2**

Monomer	Repeating unit of the polymer
Structure of the monomer used to form polymer K	Polymer K $ \begin{array}{c} \text{C}_2\text{H}_5 \quad \text{H} \\ \quad \\ -\text{C} - \text{C}- \\ \quad \\ \text{H} \quad \text{H} \end{array} $
Monomer L 	Repeating unit of the polymer formed from monomer L

0 4 . 2

State why polymer **K** in Question **04.1** is not biodegradable.**[1 mark]**



0 4 . 3

Table 3 shows data from the analysis of a hydrocarbon.

Table 3

Element	% by mass
Carbon	88.9
Hydrogen	11.1

The mass spectrum of the hydrocarbon gives $M_r = 54.0$

Calculate the empirical formula **and** the molecular formula of the hydrocarbon.
Show your working.

[3 marks]

Empirical formula _____

Molecular formula _____

6

Turn over ►



0 5

This question is about rates of reaction of halogenoalkanes.

0 5 . 1

Define rate of reaction.

[1 mark]

A student investigates the reactivity of halogenoalkanes.

- The student adds a halogenoalkane to ethanol and sodium hydroxide solution in a test tube.
- The student warms the mixture in the test tube at approximately 60 °C
- The halogenoalkane is hydrolysed.

0 5 . 2

Suggest how the student should heat the test tube safely.

[1 mark]

Nitric acid and silver nitrate solution are added to the test tube in Question **05.2** to form a precipitate.

The time for a precipitate to appear is recorded.

The experiment is repeated with other halogenoalkanes.

The rate of reaction for each halogenoalkane is calculated and the results are shown in **Table 4**.

Table 4

Halogenoalkane	Rate of formation of precipitate / s ⁻¹
1	0.0042
2	0.0054
3	0.0087

0 5 . 3

State the halogenoalkane that reacts in the shortest time.

[1 mark]



Samples of 1-chloropropane, 1-bromopropane and 1-iodopropane are tested in this experiment.

0 5 . 4

Suggest why 1-iodopropane has the fastest reaction rate.

[1 mark]

0 5 . 5

Explain why these reactions are slower at a lower temperature. You must refer to particles in your answer.

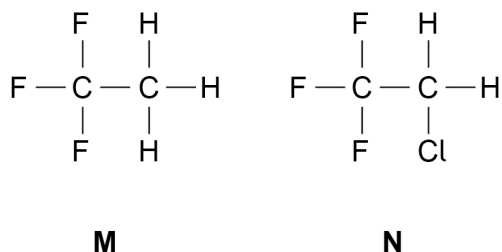
[2 marks]

6

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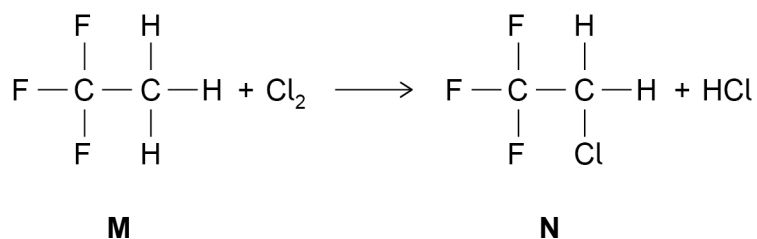
0 6Compounds **M** and **N** are gaseous organic compounds.**0 6 . 1**Calculate the percentage by mass of fluorine in **M** ($M_r = 84.0$).
Give your answer to 3 significant figures.**[1 mark]**

Percentage by mass of fluorine _____

0 6 . 2Suggest why **M** does not damage the ozone layer.**[1 mark]**



M can be converted to **N** in the presence of chlorine and ultraviolet light.



0 6 . 3 Name the mechanism for this reaction.

[1 mark]

0 6 . 4 2.64 g of **M** ($M_r = 84.0$) are reacted with chlorine.
The percentage yield of **N** is 93.0%

Calculate the mass, in g, of **N** formed in this reaction.

[3 marks]

Mass of **N** _____ g

0 6 . 5 A sample of **M** and chlorine are reacted in a sealed 10 dm³ container at a given temperature and in the presence of ultraviolet light.

State why the rate of reaction will increase if the reaction is repeated in a 5 dm³ container.

[1 mark]

Turn over ►



0	7
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This question is about alkenes and halogenoalkanes.

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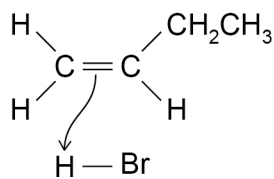
Alkenes react with hydrogen halides to produce halogenoalkanes.

Name and complete the mechanism to show the major product formed in the reaction between but-1-ene and hydrogen bromide.

[4 marks]

Name of mechanism _____

Mechanism



Halogenoalkanes can react under different conditions to make products with different functional groups.

0 7 . 2

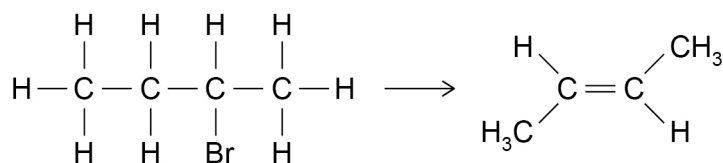
Give the meaning of functional group as applied to an organic molecule.

[1 mark]

0 7 . 3

Give the reagent and **one** condition required for the following change.

[2 marks]



Reagent _____

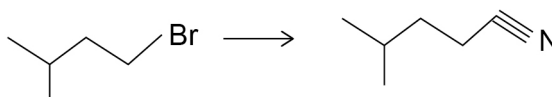
Condition _____

0 7 . 4

Give the reagent and **one** condition required for the following change.

Use IUPAC rules to name the organic product.

[3 marks]



Reagent _____

Condition _____

IUPAC name of product _____

10

Turn over ►

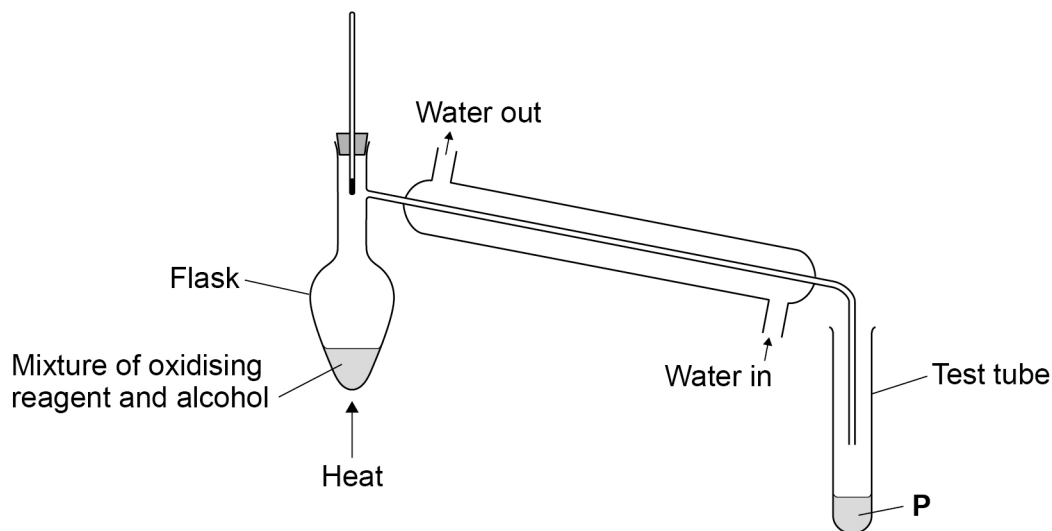


0 8

This question is about alcohols and their oxidation products.

Figure 3 shows the equipment that can be used to oxidise alcohols.

Figure 3

**0 8 . 1**

Suggest one way to increase the yield of product **P** collected in the test tube in **Figure 3**.

[1 mark]

0 8 . 2

Name the practical technique used to obtain **P**.

State the reagent (or combination of reagents) used in the flask to oxidise the alcohol.

Describe the colour change observed in this reaction.

[3 marks]

Practical technique _____

Reagent(s) _____

Colour change _____



0 8 . 3

The carboxylic acid functional group can be identified by its reaction with aqueous NaHCO_3

Give the observation when propanoic acid reacts with aqueous NaHCO_3

Write an equation for this reaction.

[2 marks]

Observation _____

Equation _____

0 8 . 4

Identify the organic product of the reaction of 2-methylbutanal with Fehling's solution.

State what is observed in this reaction.

[2 marks]

Organic product _____

Observation _____

0 8 . 5

An alcohol with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ does **not** react with the oxidising reagents used in this question.

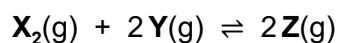
Draw the displayed formula of this alcohol.

[1 mark]



0 9

This question is about the equilibrium



0 9 . 1

A student adds 2.0 mol of X_2 to 3.0 mol of Y .

The mixture is placed in a sealed container at a given temperature to reach equilibrium.

The equilibrium mixture contains 1.5 mol of X_2 Complete **Table 5** to show the amount, in mol, of each substance at equilibrium.**[2 marks]****Table 5**

	X_2	Y	Z
Initial amount / mol	2.0	3.0	0.0
Equilibrium amount / mol	1.5		

0 9 . 2

The student repeats the experiment with different amounts of X_2 and Y at a different temperature.At equilibrium, the concentration of $\text{X}_2 = 0.150 \text{ mol dm}^{-3}$ and the concentration of $\text{Y} = 0.0520 \text{ mol dm}^{-3}$ At this temperature, the value of the equilibrium constant, $K_c = 290 \text{ mol}^{-1} \text{ dm}^3$ Write the expression for the equilibrium constant, K_c Calculate the concentration of Z , in mol dm^{-3} , at equilibrium.**[3 marks]** K_c

Calculation

Concentration of Z _____ mol dm^{-3} 

Table 6 shows the value of the equilibrium constant (K_c) at different temperatures.

Table 6

Temperature / K	$K_c / \text{mol}^{-1} \text{dm}^3$
300	780
500	160
700	55

0 9 . 3

State how the equilibrium concentration of **Z** changes as the temperature increases.

Explain your answer.

[3 marks]

Change in equilibrium concentration of **Z** _____

Explanation _____

0 9 . 4

The student repeats the experiment under different conditions.

Complete **Table 7** to state the effect on the value of equilibrium constant (K_c) and the position of the equilibrium when

- a catalyst is added
- the volume of the container is increased.

[2 marks]

Table 7

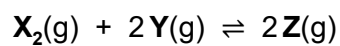
	Effect on K_c	Effect on position of equilibrium
Catalyst is added		
Volume of the container is increased		

Question 9 continues on the next page

Turn over ►



The equilibrium is repeated here.



At 300 K, the value of the equilibrium constant, $K_c = 780 \text{ mol}^{-1} \text{ dm}^3$

0 9 . 5

Calculate a value for the equilibrium constant (K_c) for the reverse reaction at 300 K
and give the units for this K_c

[2 marks]

K_c _____

Units _____

12

END OF QUESTIONS



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